

Zhiyun Guo

gzy7777@sjtu.edu.cn github.com/gggzy1

RESEARCH INTERESTS

- **Optimization:** First-order methods, stochastic optimization, and large-scale optimization.
- **EconCS (Computational economics):** Revealed preference, market equilibrium, and algorithmic approaches to learning and mechanism design in economic systems.

EDUCATION

Undergraduate, Shanghai Jiao Tong University Shanghai, China
School of Ocean and Civil Engineering 2022 – 2023
Antai College of Economics and Management 2023 – 2027 (Expected)

- **Major:** Big Data Management and Applications (Big Data and Management Science)
- **Minor:** Computer Science and Technology
- **Major GPA:** 90.95/100
- **Related Courses:** Advanced Probability and Statistics (PhD level, 91/100), Management Modeling and Optimization (96/100), Optimization Methods in Economics (94/100), Regression Analysis (96/100), Econometrics (95/100), Machine Learning (94/100), C++ (92/100), Data Structures (92.5/100), Linear Algebra (100/100)

RESEARCH EXPERIENCE

Revealed preferences learning Dec 2025 – Now
Research Assistant, Shanghai Jiao Tong University
(supervised by Dr. Chuwen Zhang, Prof. Zizhuo Wang and Prof. Yinyu Ye)

- <https://arxiv.org/abs/2606.23985>.
- Developed a computationally tractable surrogate-market framework for learning collective revealed preferences, with surrogate consumers selected via *Frank-Wolfe algorithm*.
- Derived generalization bounds for multiple classes of surrogate consumers.
- Designed and implemented synthetic experiments in PyTorch, including ablations over Frank-Wolfe variants, step-size strategies, problem scales, and neural-network baselines.
- Submitted to *ICLR2027*. Extended manuscript in preparation.

Stochastic optimization Mar 2026 – Now
Research Assistant, Shanghai Jiao Tong University (supervised by Prof. Qi Deng)

- Studied adaptive stochastic approximation methods for two- and multi-stage stochastic programs.
- Implemented and evaluated the algorithms on stochastic programming benchmarks.
- Conducted extensive computational comparisons with Dynamic Stochastic Approximation (DSA) and related baselines.
- *Manuscripts in preparation*

Tuning-Free First-Order Algorithms Apr 2025 – Jan 2026
Research Assistant, Shanghai Jiao Tong University (supervised by Prof. Qi Deng)

- Designed parameter-free algorithm for non-smooth convex optimization that requires no prior knowledge of Lipschitz constants or initial distance to optimum.
- Proved convergence guarantees across three settings: deterministic non-smooth, stochastic non-smooth with high-probability bounds via martingale concentration, and stochastic smooth.
- Validated empirically on L_1 regression and SVM benchmarks, demonstrating convergence improvements over baselines in both iteration and wall-clock time.
- *Manuscript in preparation.*

Hydropower Reservoir Scheduling

May 2025 – Feb 2026

Research Assistant, Shanghai Jiao Tong University (supervised by Prof. Qi Deng)

- Reformulated an industry hydropower scheduling model as a risk-averse multi-stage stochastic program by incorporating CVaR constraints, capturing tail-risk in uncertain inflow scenarios.
- Experimented with multi-stage stochastic programming solvers, e.g., SDDP(Stochastic Dual Dynamic Programming) and SDDiP(Stochastic Dual Dynamic Integer Programming).
- Contributed to the paper: <https://www.mdpi.com/2073-4441/18/10/1220>.

Large Language Models for Operations Research

Jul 2024 – Mar 2025

LLM intern, Cardinal Operations

- Participated in fine-tuning and evaluation of LLMs for operational research tasks.

COURSE PROJECTS

Algorithmic Collusion in Recommendation Platform Competition

Dec 2024 – Feb 2025

Research Project, Shanghai Jiao Tong University (supervised by Prof. Jialu Liu)

- Developed a duopoly model for competing recommendation platforms, incorporating linear demand functions across traditional and recommendation-driven markets.
- Formulated Nash and monopoly equilibria and implemented a multi-agent Q-learning framework in Python to simulate platform interactions and strategic decision-making.
- Evaluated algorithmic convergence across diverse pricing regimes, identifying emergent supra-Nash collusion in monopoly settings without explicit coordination.

TEACHING EXPERIENCE & VOLUNTARY EXPERIENCE

Linear Algebra Academic Tutor

Dec 2025 – Jan 2026

Shanghai Jiao Tong University, Antai College of Economics and Management

- Delivered two review sessions and assisted students with course-related questions.

Stepping Stones Shanghai Program (Digital Literacy)

May 2024

- Taught three online visual-programming sessions to underserved children in China.

HONORS & AWARDS

- The Chinese Mathematics Competitions for College Students (CMC), First Prize in Shanghai 2024
- Shanghai Municipal Scholarship 2024
- Undergraduate Academic Excellence Scholarship 2023, 2025

TECHNICAL SKILLS

- **Programming:** Julia, Python, C++, Matlab, SQL.
- **Languages:** English (Fluent), Mandarin (Native).

HOBBIES & OTHER INTERESTS

- Music: piano.
- Outdoor activities: jogging, swimming.